

CS 4530: Fundamentals of Software Engineering

Module 10: Testing Larger Things

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Learning Objectives for this Lesson

- By the end of this lesson, you should be prepared to:
 - Design test cases for code using fakes, mocks and spies
 - Explain why you might need a test double in your testing
 - Explain why you might need tests that are larger than unit tests
 - Explain how large, deployed systems lead to additional testing challenges

Why do we test?

- Unit Testing
 - Does the SUT satisfy its specification?
- Integration Testing
 - Do the SUT and its context work correctly *together*?
- Acceptance Testing
 - Does the SUT satisfy the customer
 - “Good” test suite answers: Are we building the right system ?

Unit Testing

What does it mean for a unit test to succeed?

- *Test Oracles* define the criteria for a test to succeed.

Possible kinds of test oracles

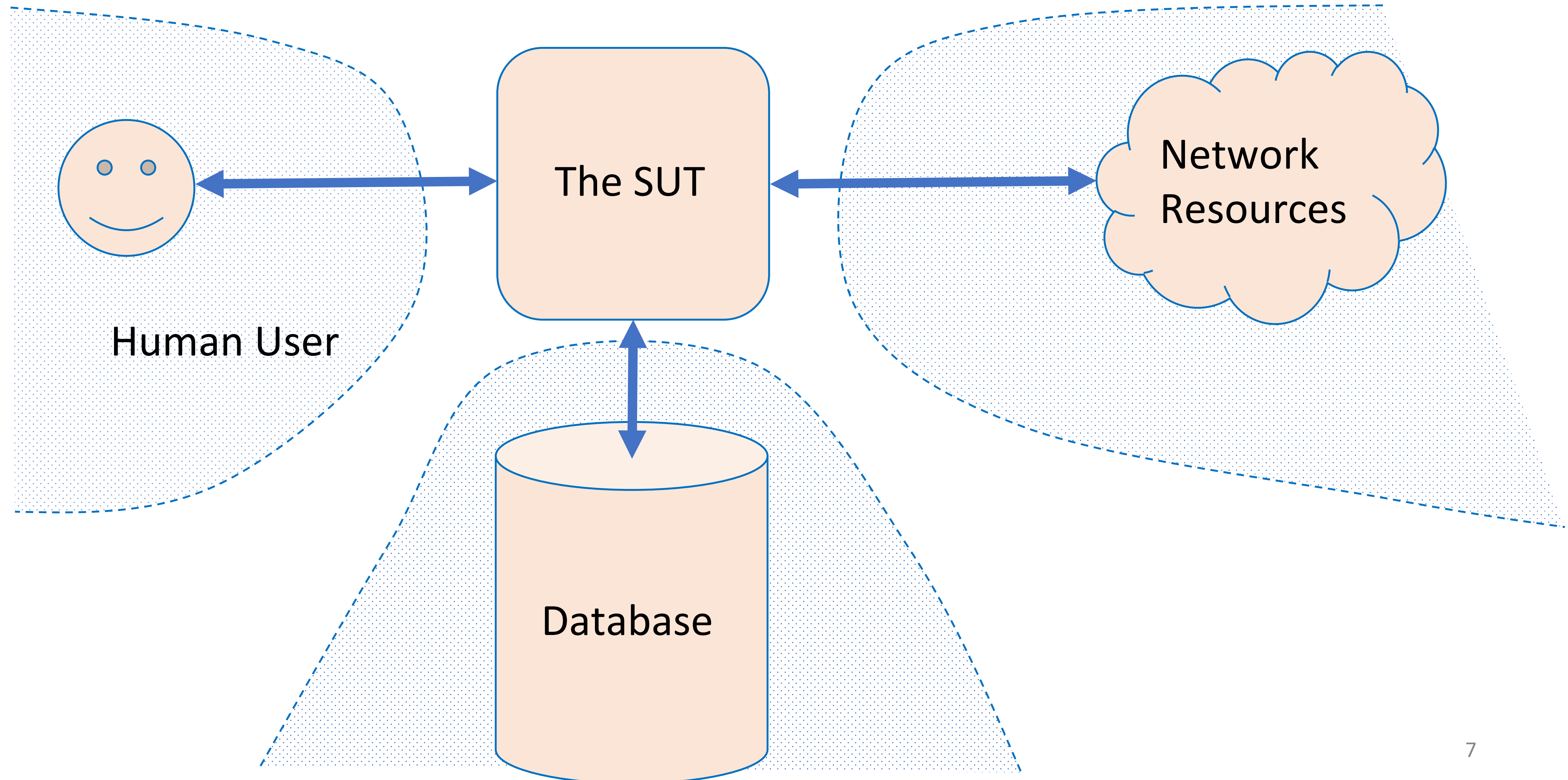
- Function returns the exact “right” answer
- Function returns an acceptable answer
- Returns the same value as last time
- Function returns without crashing
- Function crashes (as expected)
- Function has the right effects on its environment



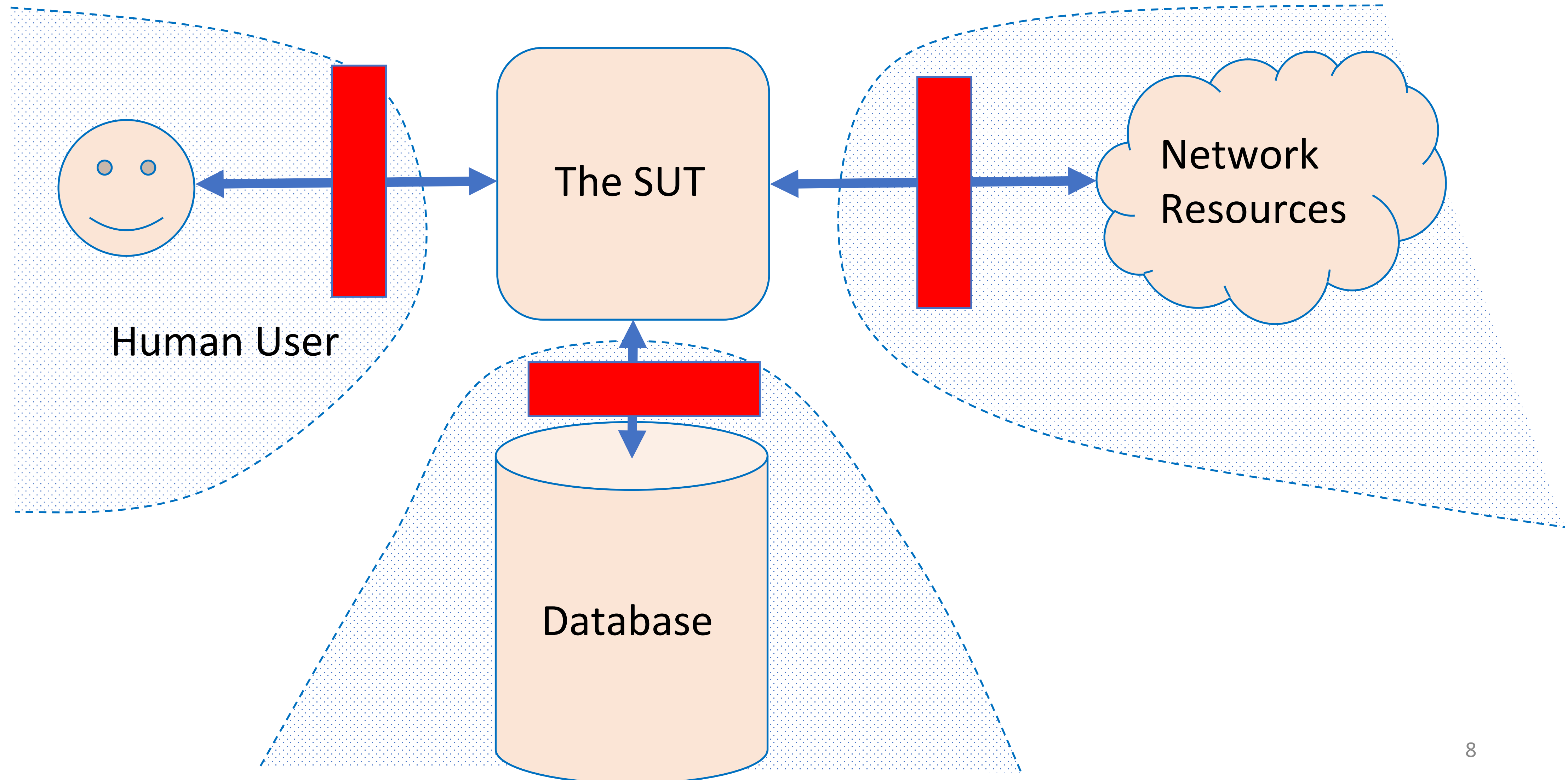
Story so far: Tests Check Return Values

```
test('addStudent should add a student to the database', () => {  
  // const db = new DataBase ()  
  expect(db.nameToIDs('blair')).toEqual([])  
  
  const id1 = db.addStudent('blair');  
  
  expect(db.nameToIDs('blair')).toEqual([id1])  
});
```


But our functions don't exist in isolation



Test doubles replace uncontrollable things with things that you do control



“Test Doubles” Stand In For Other Components

- Act as a stand-in for components, allowing for testing in isolation
- Fakes: Replace client implementations with dummies for testing
- Mocks: Automatically-generated fake implementations for an interface
- Spies: Automatically-instrument internals of objects, classes or modules

Example: How to test the ProducerClock?

```
export interface IClockWithListeners {  
  reset():void // resets the time to 0  
  tick():void // increment time and notify all listeners  
  // add a listener and initialize it with the current time  
  addListener(listener:IClockListener):void  
}
```

10

```
export interface IClockListener {  
  // @param t - the current time, as reported by the clock  
  notify(t:number):void  
}
```

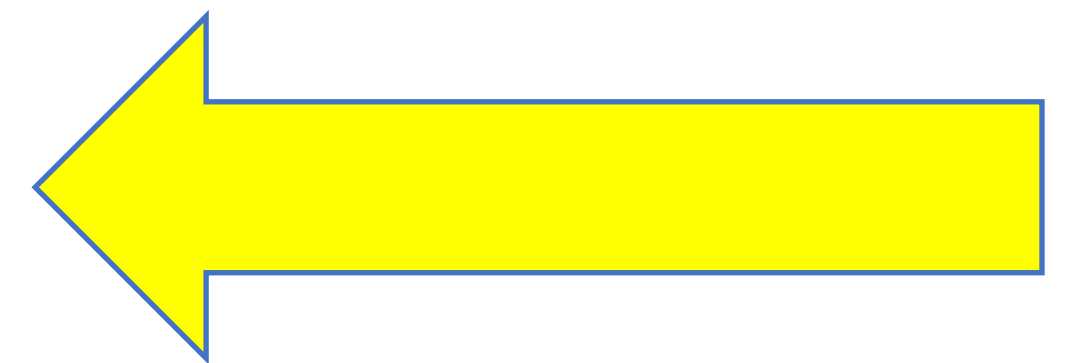
```
export class ProducerClock implements IClockWithListeners {  
  // some implementation  
}
```

You could test the Producer Clock with a hand-built test double (a "fake")

src/clockWithObserverPattern.spec.ts

```
export interface IClockWithListeners {  
  reset():void // resets the time to 0  
  tick():void // increment time and notify all listeners  
  // add a listener and initialize it with the current time  
  addListener(listener:IClockListener):void  
}
```

```
class ClockListenerForTest implements IClockListener {  
  private _time : number = 0  
  constructor (masterClock:IClockWithListeners) {  
    masterClock.addListener(this)  
  }  
  notify (t:number) : void {this._time = t}  
  getTime () : number {return this._time}  
}
```



Now we can test using the fake observer

```
import { ProducerClock } from "../clockWithObserverPattern";

const clock1 = new ProducerClock
const listener1 = new ClockListenerforTest(clock1)

describe("tests for ProducerClock", () => {
  test("after reset, listener should return 0", () => {
    clock1.reset()
    expect(listener1.getTime()).toBe(0)
  })
  test("after one tick, listener should return 1", () => {
    clock1.reset(); clock1.tick()
    expect(listener1.getTime()).toBe(1)
  })
  test("after two ticks, listener should return 2", () => {
    clock1.reset(); clock1.tick(); clock1.tick()
    expect(listener1.getTime()).toBe(2)
  })
})
```

Does using the fake listener solve the problem?

```
class ClockListenerForTest implements
IClockListener {
    private _time : number = 0
    constructor (private
masterClock:IClockWithListeners) {
        masterClock.addListener(this)
    }
    notify (t:number) : void
    {this._time = t}
    getTime () : number {return
this._time}
}
```

- Good news:
 - It works!
 - It doesn't require learning other libraries
- Bad news:
 - It's a maintenance burden (what if new methods are added to IClockListener?)
 - It took manual effort to write
 - Richer fakes (e.g. track how many times a method called) would take even more effort to write

Mocks are automated fakes

- Vitest's mocks return "undefined" by default (can be customized), and track calls to the function

```
test("simplest mock behavior", () => {  
  const mockFunction1 = vi.fn();  
  
  const result1 = mockFunction1("17");  
  const result2 = mockFunction1("42")  
  
  expect(result1).toBeUndefined();  
  expect(result2).toBeUndefined()  
  
  expect(mockFunction1).toHaveBeenCalled();  
  expect(mockFunction1).toHaveBeenCalledTimes(2);  
  
  expect(mockFunction1).toHaveBeenCalledWith("17");  
  expect(mockFunction1).toHaveBeenCalledWith("42")  
});
```


You can customize your mock in many ways

```
test("customizing mock functions", () => {  
  // you can specify the the return value  
  const mockFunction3 = vi.fn();  
  mockFunction3.mockReturnValue("baz");  
  expect(mockFunction3(17)).toBe("baz");  
  expect(mockFunction3).toHaveBeenCalledWith(17);
```

<https://vitest.dev/guide/mocking>

```
  // or give the mock an implementation  
  const mockFunction2 = vi.fn()  
  mockFunction2.mockImplementation((n: number) => n + n);  
  expect(mockFunction2(3)).toBe(6);  
  expect(mockFunction2(14)).toBe(28)  
  expect(mockFunction2).toHaveBeenCalledWith(3);  
  expect(mockFunction2).toHaveBeenCalledWith(14);
```

MockReset erases history; returns
implementation to 'undefined'

```
  // you can also reset the mock's history and implementation  
  mockFunction2.mockReset()  
  expect(mockFunction2).not.toHaveBeenCalledWith(14);  
});
```

You can mock classes and interfaces using vitest-mock-extended

src/vitest-mock-extended.spec.ts

```
import { describe, test, expect, beforeEach } from 'vitest';
import { mock, mockClear } from 'vitest-mock-extended';
import { type IClockListener, ProducerClock } from './clockWithObserverPattern';
```

```
const clock1 = new ProducerClock();
const listener1 = mock<IClockListener>();
clock1.addListener(listener1);
```

Construct a mock listener
(1 line!)

```
describe('tests for ProducerClock', () => {
  beforeEach(() => {
    mockClear(listener1); //Clear the mock function's history, but leave the mock implementation as is
  });
  test('after one tick, listener should have been notified with 1', () {
    clock1.reset(); clock1.tick();
    expect(listener1.notify).toHaveBeenLastCalledWith(1);
  });
  test('after two ticks, listener should have been notified with 1 and 2', () => {
    clock1.reset(); clock1.tick(); expect(listener1.notify).toHaveBeenLastCalledWith(1);
    clock1.tick(); expect(listener1.notify).toHaveBeenLastCalledWith(2);
    expect(listener1.notify).toHaveBeenCalledTimes(2);
  });
});
```

All the methods of
IClockListener are mocked.

<https://www.npmjs.com/package/vitest-mock-extended>

Unlike mocks, spies *instrument* existing implementations

- Consider cases where you *don't* want a complete fake, but *do* want to check side-effects:
 - What was sent on the network?
 - How many times was a problem logged?
 - What was inserted in the database?
- Vitest can automatically instrument existing code to make it into a “spy” – a mock but with the original implementation

Spy
"remembers"

Real
implementation
is used

Use vi.spyOn to create a spy

src/clockWithSpy.spec.ts

```
import { describe, test, expect, beforeEach, vi } from 'vitest';
import { ClockListener, ProducerClock } from '../clockWithObserverPattern';

const clock1 = new ProducerClock();
const clockClient = new ClockListener(clock1);
const notifySpy = vi.spyOn(clockClient, 'notify'); // Spy on calls to notify on this clock
describe('tests for ProducerClock', () => {
  beforeEach(() => {
    notifySpy.mockClear(); // Clear the mock function's history
  });
  test('after one tick, listener should return 1', () => {
    clock1.reset(); clock1.tick();
    expect(notifySpy).toHaveBeenLastCalledWith(1);
  });
  test('after two ticks, listener should return 2', () => {
    clock1.reset(); clock1.tick();
    expect(notifySpy).toHaveBeenLastCalledWith(1);
    clock1.tick();
    expect(notifySpy).toHaveBeenLastCalledWith(2);
    expect(notifySpy).toHaveBeenCalledTimes(2);
  });
});
```

Spies can be used even when you can't control the SUT

- You can specify *any* object, and *any* method name (even private methods)
- Spy on objects *or* entire modules
- The spy logs *all* calls to that method of that object or module
- The call to the **original still gets made**, unless the spy explicitly supplies a substitute
 - we'll illustrate this a few slides from now.

Let's use a spy to test a little http client.

```
import axios from "axios";
```

```
export class Echo {
```

```
  /**
```

```
    * @argument a string
```

```
    * @returns a promise to return the same string
```

```
    * @requires axios
```

```
    * @calls https://httpbin.org/get?answer=${str}
```

```
  */
```

```
  public static async echo(str: string): Promise<string> {
```

```
    const res = await axios.get(`https://httpbin.org/get?answer=${str}`);
```

```
    return res.data.args.answer;
```

```
  }
```

```
}
```

src/EchoClass.ts

We'll have two tests

src/EchoClass.spec.ts

```
import { Echo } from './EchoClass';
import axios from 'axios';
import { describe, it, expect, vi } from 'vitest'

describe('tests for Echo.echo', () => {

  it('should return the same string it receives', async () => {
    const str = 'test';
    const res = await Echo.echo(str);
    expect(res).toEqual(str);
  });

  it('should call axios.get', async () => {
    const spy1 = vi.spyOn(axios, 'get');
    const str = 'test';
    const res = await Echo.echo(str);
    expect(spy1).toHaveBeenCalled();
    expect(spy1).toHaveBeenCalledTimes(1);
    spy1.mockRestore();
  });
});
```

What if we wanted to test a client of echo?

- But we didn't want to issue any http requests

src/echoClient.ts

```
import { Echo } from './EchoClass';

/** calls echo twice and concatenates the results */
export async function echoClient(str: string) {
  const res1 = await Echo.echo(str);
  const res2 = await Echo.echo(str);
  return res1 + res2;
}
```


Solution: create a mock for Echo.echo

```
import { echoClient } from './echoClient';
import { Echo } from './EchoClass';
import axios from 'axios';
import { describe, it, expect, vi, beforeEach } from 'vitest';

describe('tests for echoClient', () => {
  beforeEach(vi.resetAllMocks);
  beforeEach(() => {
    // mock axios.get to always throw an error;
    // if any of our tests call the real axios.get, they will fail with the error
    vi.spyOn(axios, 'get')
      .mockImplementation(() => {
        throw new Error('axios.get should not be called');
      });
  });

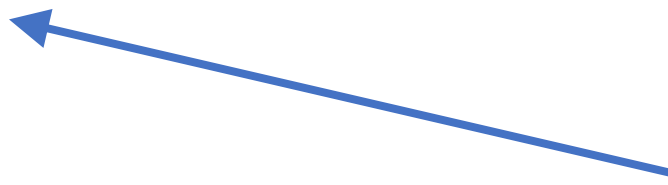
  it('echoClient should return its argument twice', async () => {
    const str = 'ABC';
    vi.spyOn(Echo, 'echo').mockImplementation(async (str: string) => Promise.resolve(str));
    const res = await echoClient(str);
    expect(res).toEqual(str + str);
  });
});
```

src/echoClient.spec.ts

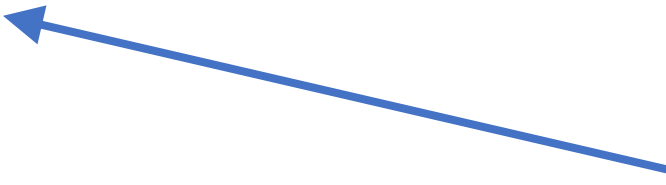
Test Doubles Have Weaknesses

- Some failures may occur purely at the integration between components:
 - The test may assume wrong behavior (wrongly encoded by mock)
 - Higher fidelity mocks can help, but still just a snapshot of the real world
- Test doubles can be brittle:
 - Spies expect a particular usage of the test double;
 - The test is "brittle" because it depends on internal behavior of SUT;
- Potential maintenance burden: as SUT evolves, mocks must evolve.

Did we correctly model the behavior of httpbin?



Not just its IO behavior, but also its dependencies



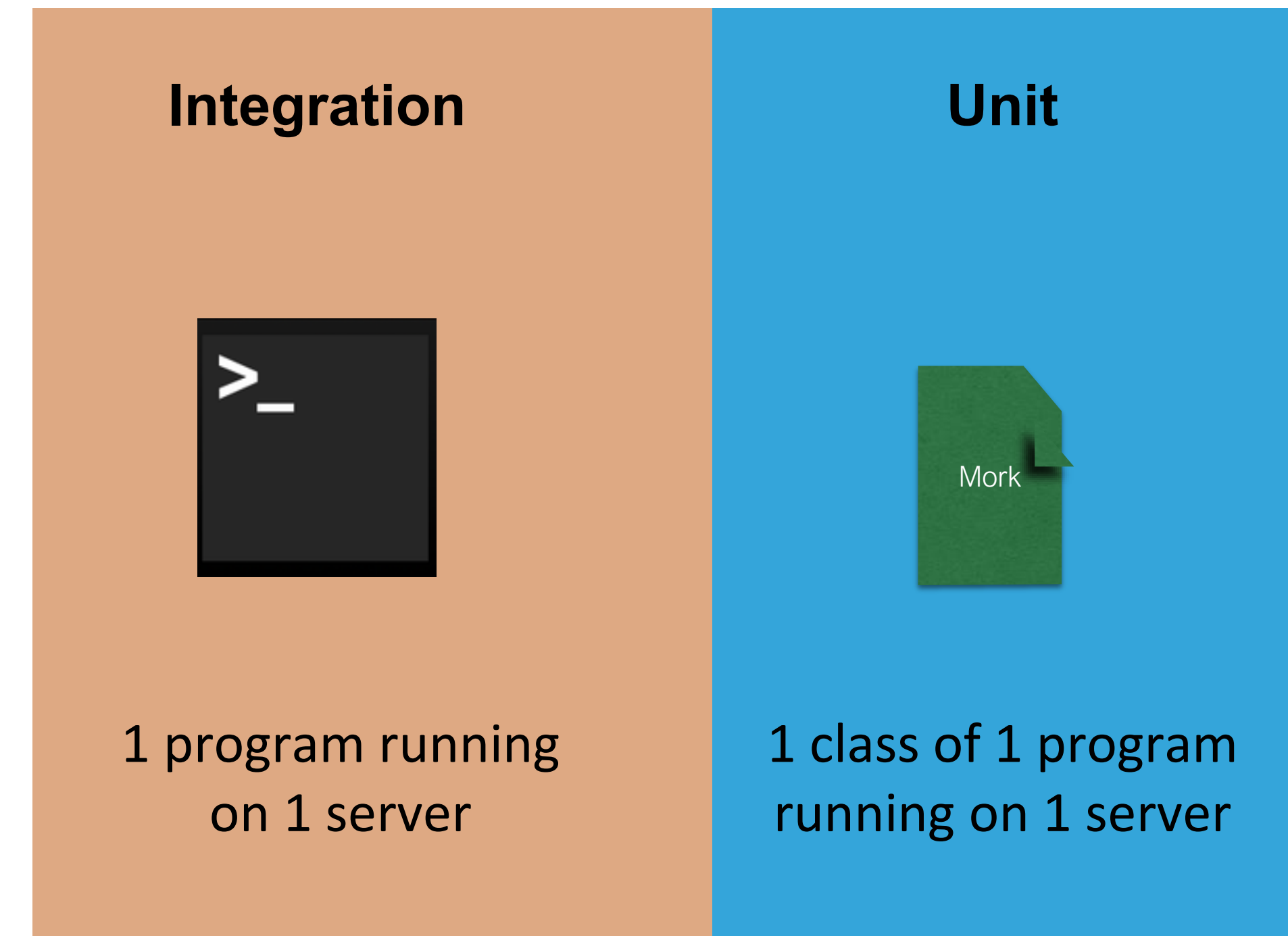
What if we didn't want to make assumptions about how httpbin behaves?

- We'd need to actually call httpbin.
- This is no longer a unit test; it's an integration test
- Which brings us to our next topic.

Integration Testing

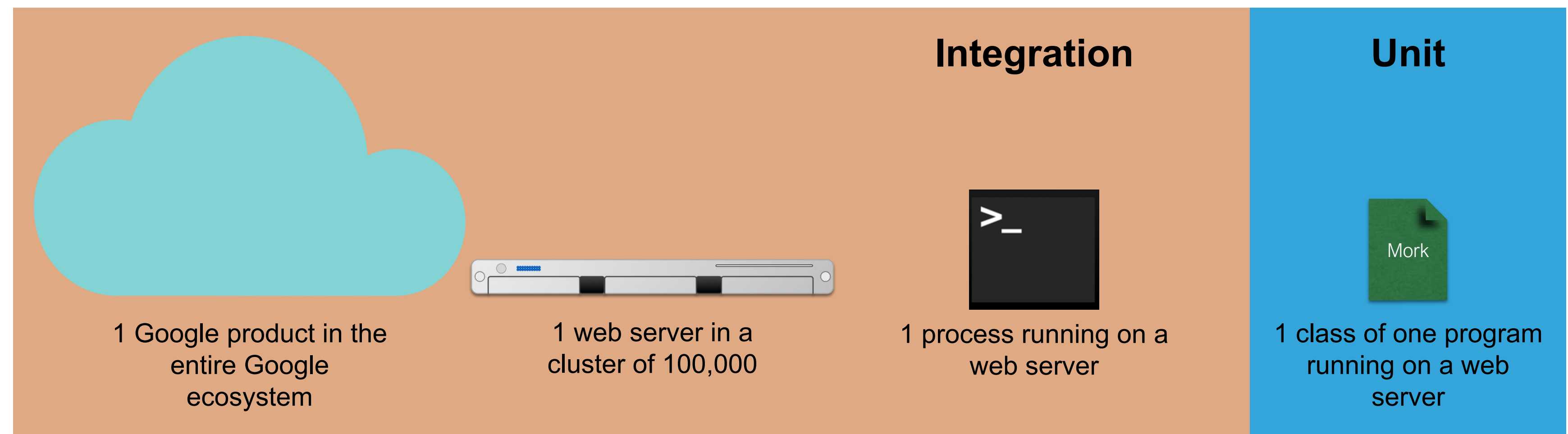
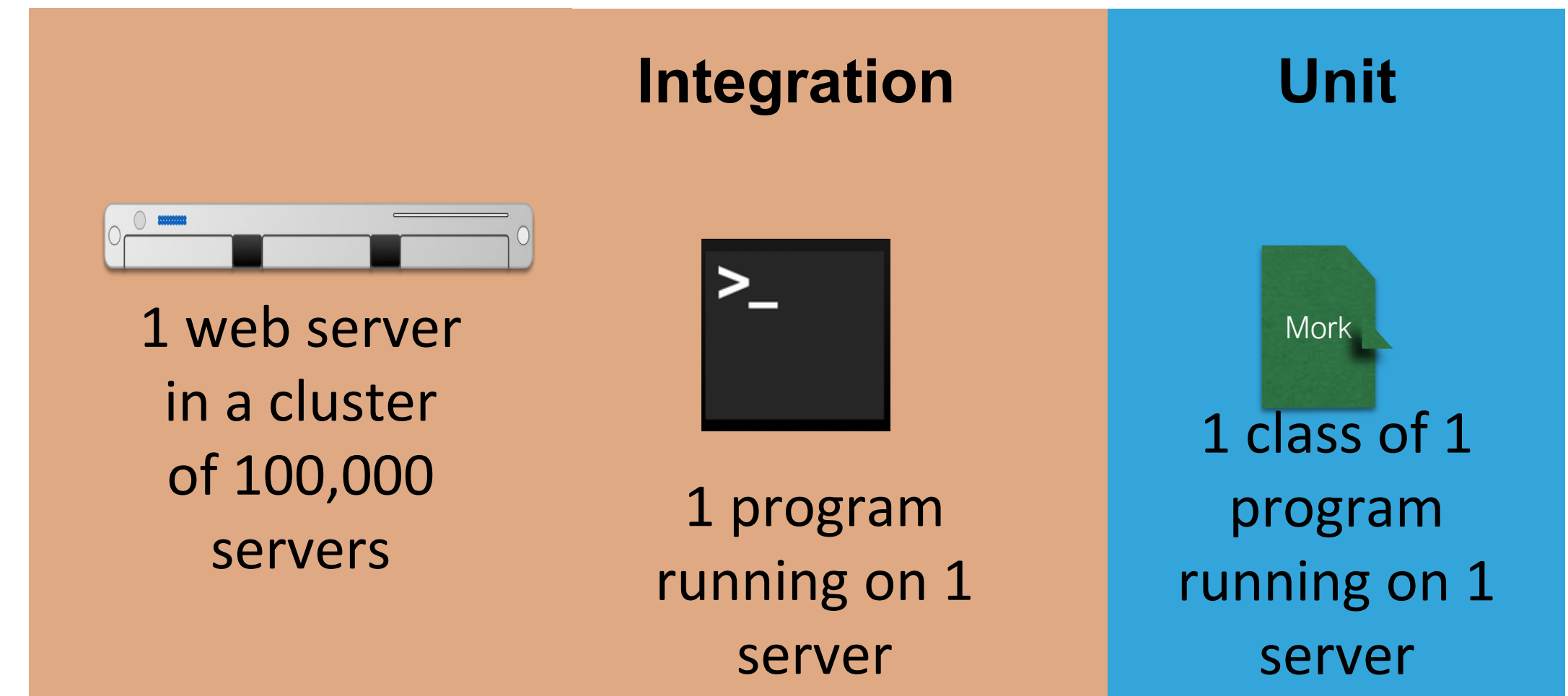
But some bugs are observable only when multiple components interact.

- These are usually because one module has made incorrect assumptions about some other module
- Unit tests won't reveal such bugs
- Mocks won't help, either (since they may incorporate our incorrect assumptions)
- So you really need integration tests

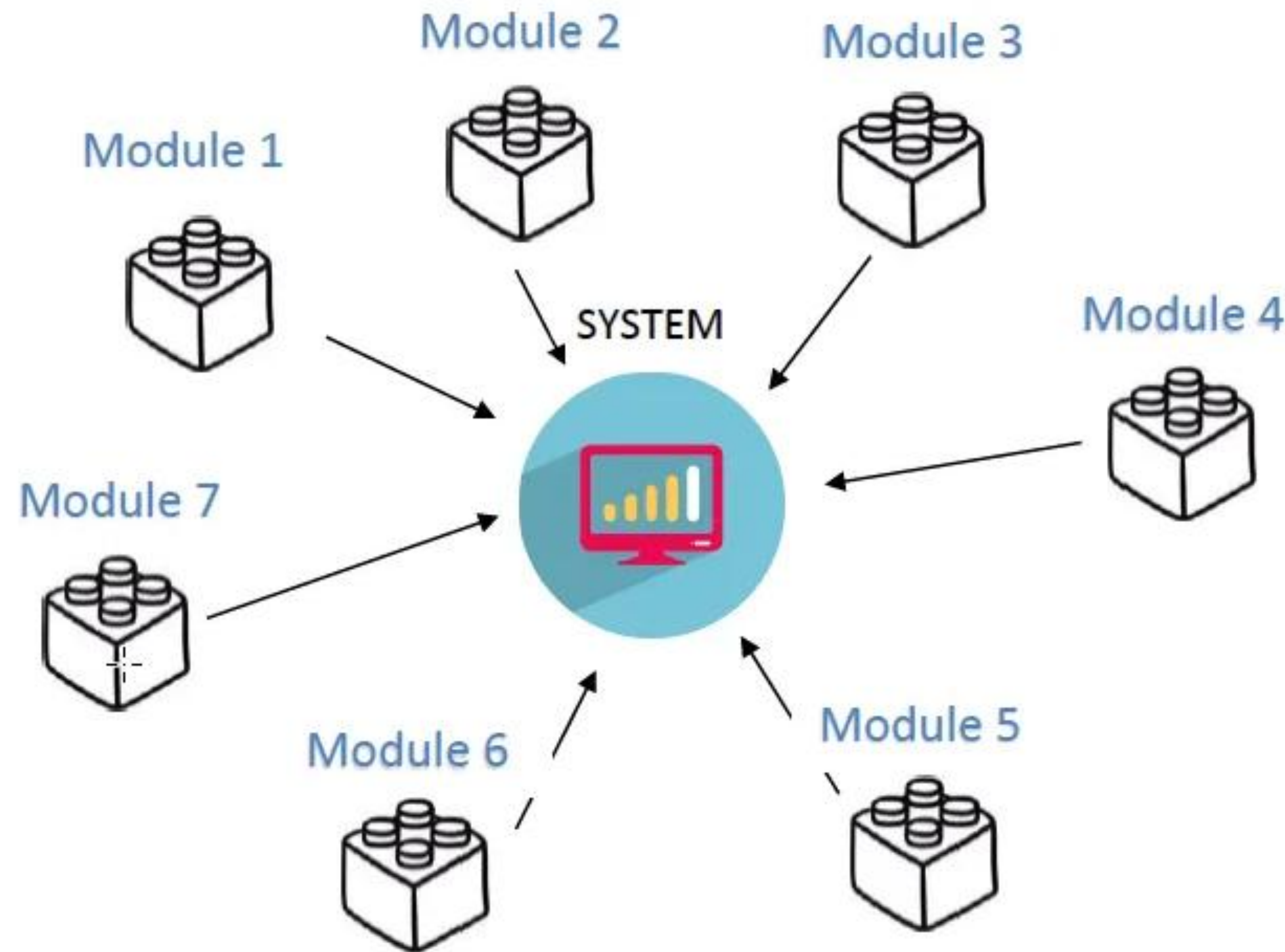


Integration tests may be larger, even enormous

- Does the presence of other jobs on our server change the behavior of our program?
- Does the presence of the other servers change the behavior of our program?

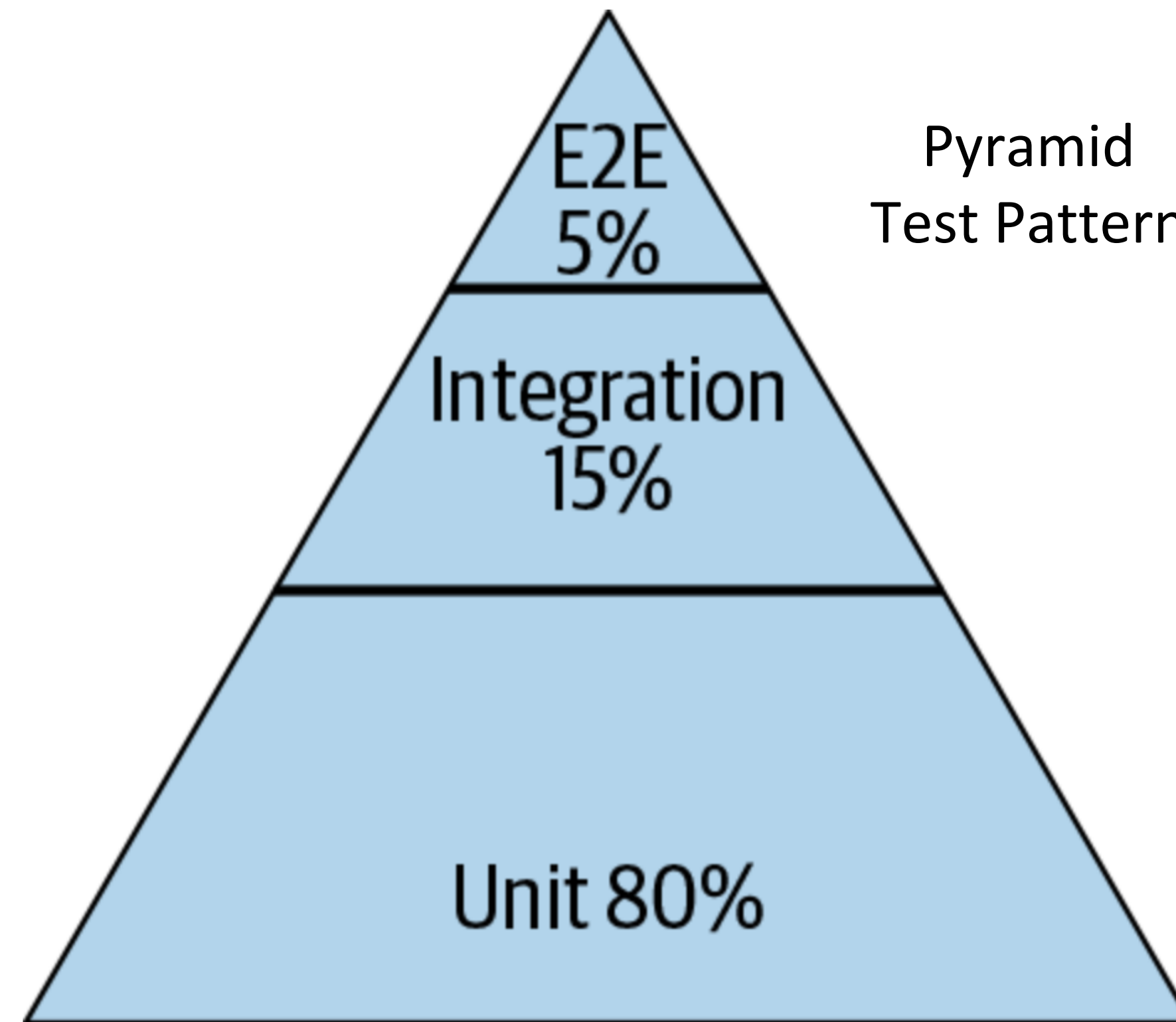
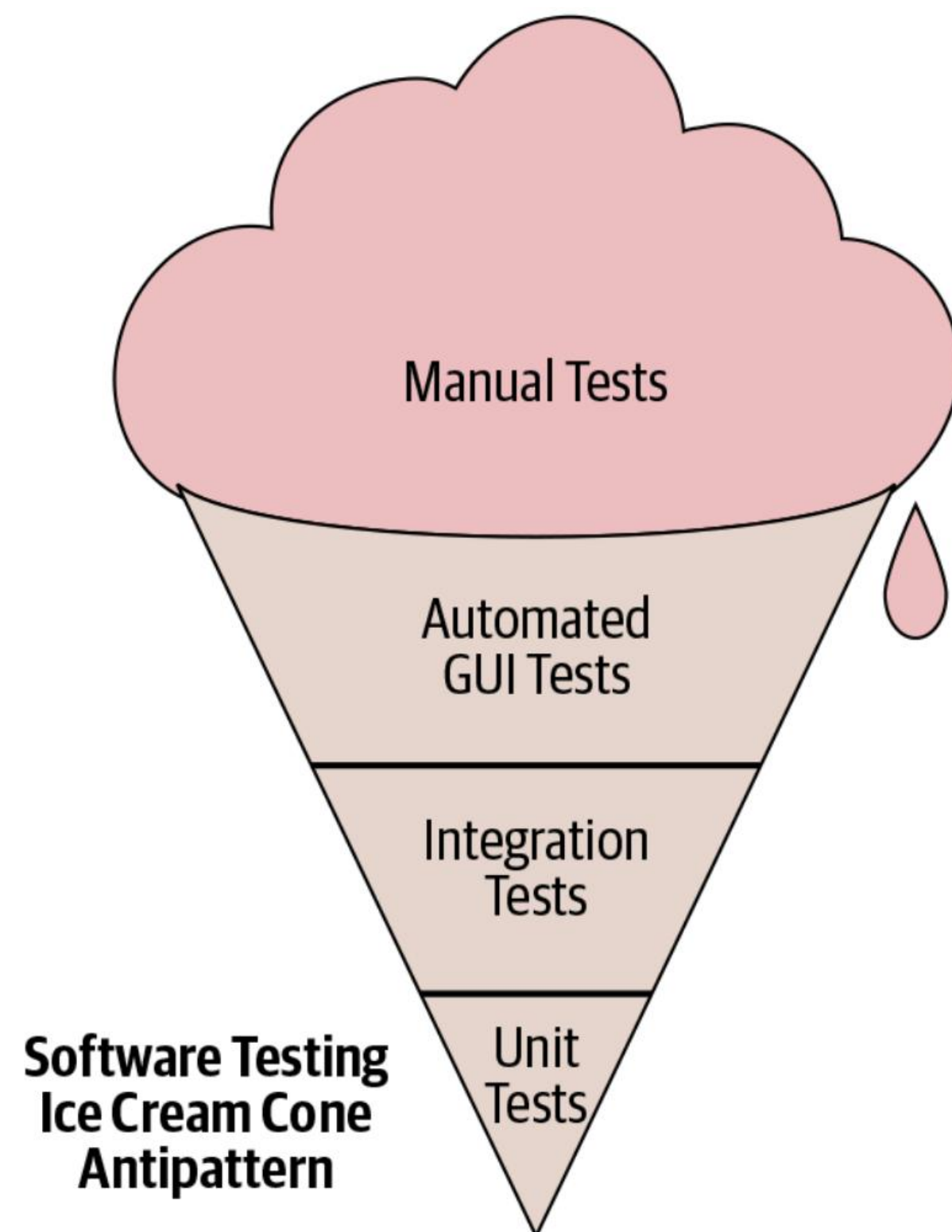


Integration tests can be done in many ways



- All at once ("Big Bang")
- Top-down
- Bottom-up
- Middle-out
- Top-Bottom-Middle
- etc., etc., etc.

Testing Distribution (How much of each kind of testing we should do?)



From SoftEng @ Google Chapter 11

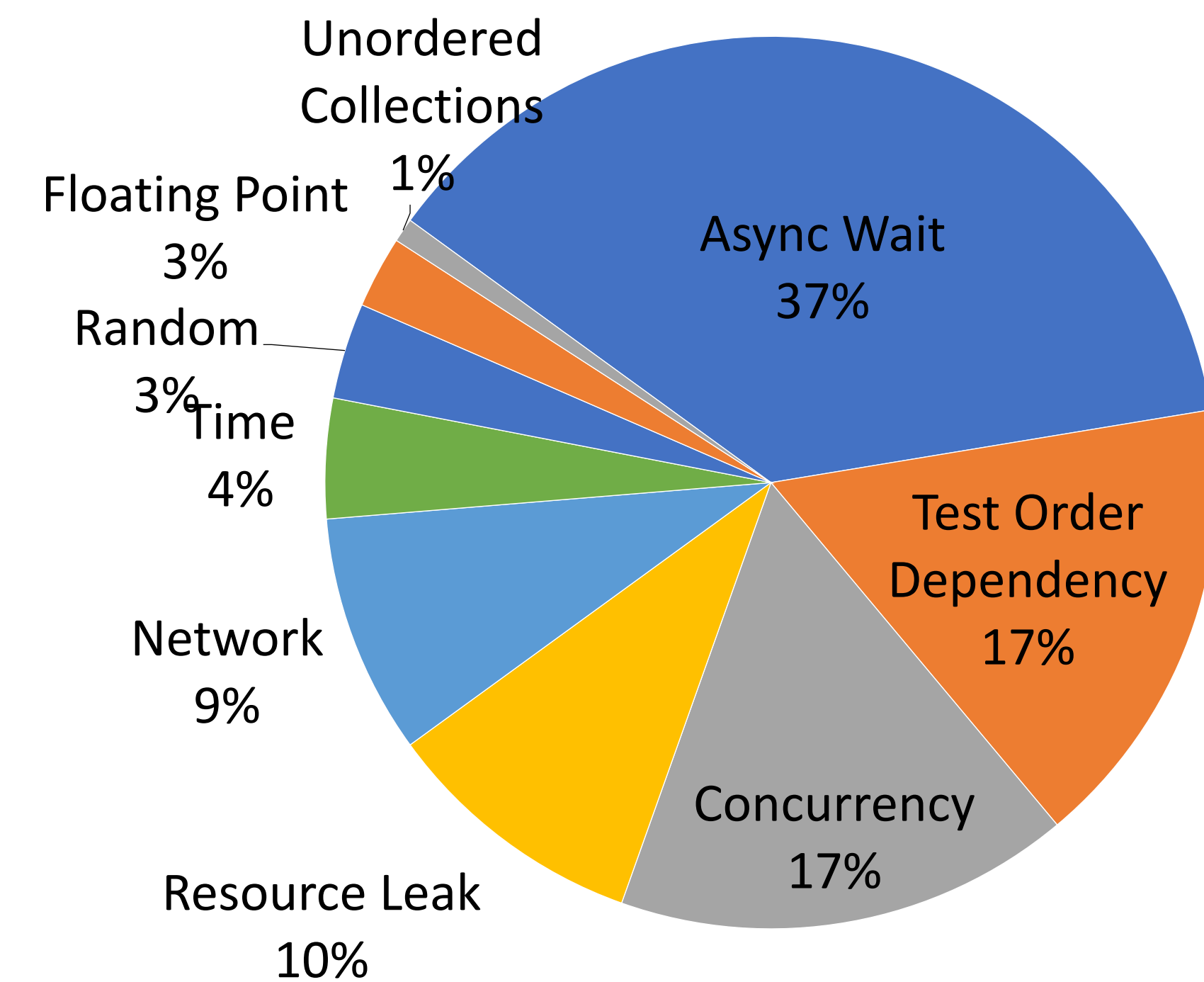
- https://learning.oreilly.com/library/view/software-engineering-at/9781492082781/ch11.html#testing_overview

How big is my test? Google's Classification

- Small: run in a single thread, can't sleep, perform I/O or make blocking calls
- Medium: run on single computer, can use processes/threads, perform I/O, but only contact localhost
- Large: Everything else

Integration Tests can be Flaky

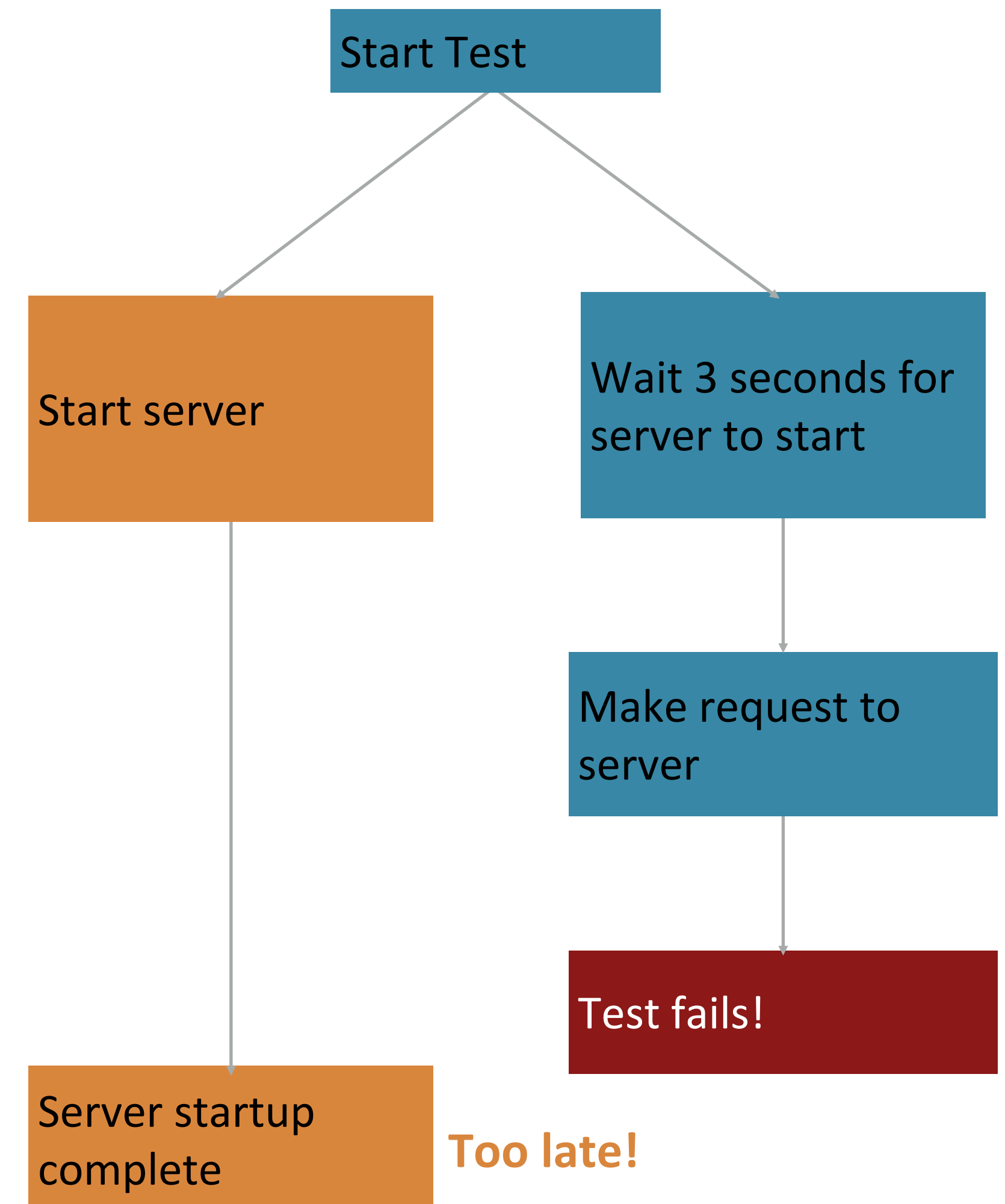
- Flaky test failures are false alarms
- Most common cause of flaky test failures:
“async wait” - tests that expect some asynchronous action to occur within a timeout
- UI Testing is often flaky and slower
- Good tests avoid relying on timing
- Good tests avoid relying on the order in which the tests are run



[Luo et al, FSE 2014 “An empirical analysis of flaky tests”]

Flaky Test Example: Async/Wait

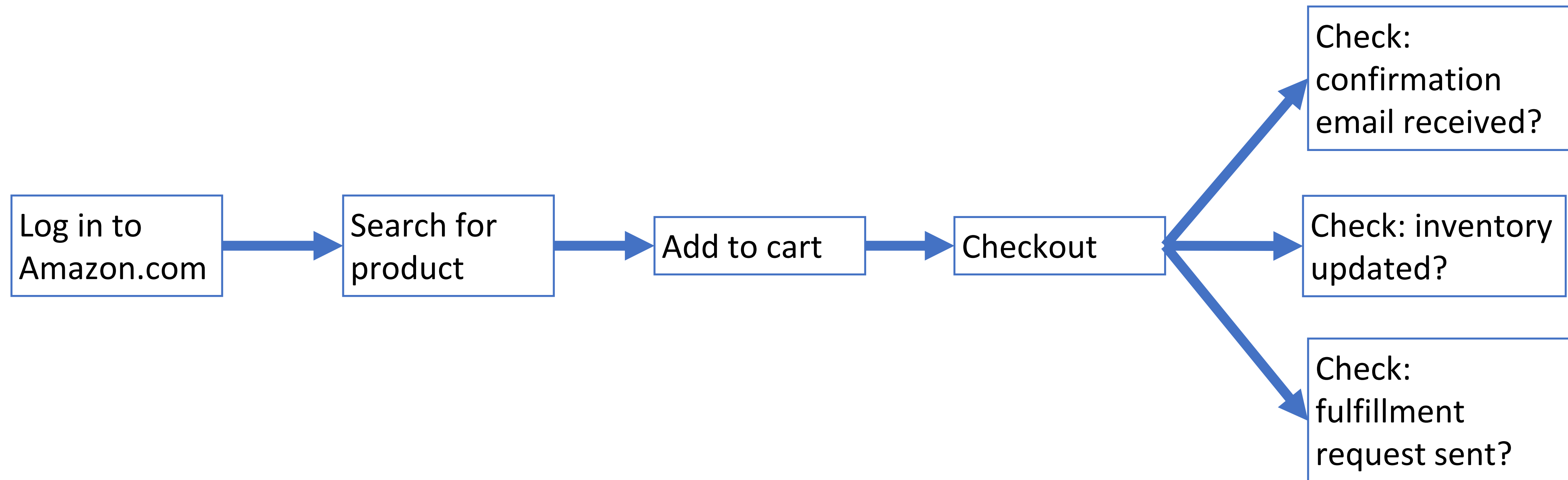
- Most common root cause of flakiness
- Difficult to avoid, but there are mitigations:
 - Have more “small” tests that don’t require concurrency
 - Ensure sufficient resources available for running tests
 - Embed reasonable error detection to classify test failures as likely to be “flaky” vs true failures



Avoiding the GUI can help reduce flakiness

- GUI makes your tests slow.
- To help reduce flakiness:
 - find a way to fire real HTTP requests without the browser (e.g., supertest library)
 - actual dependencies instead of mocks
 - Setup the test data before every test

“End-to-End” Tests can be Enormous



- Most effective end-to-end tests focus on high value user interactions (UI Testing)

Acceptance Testing

Acceptance Tests can be formulated as scenarios

- Acceptance tests are written to verify behavior from a user's perspective.
- The focus is on treating the application as a black-box
- Tests may be specified as **given-when-then scenarios**:

given there's a logged in user
and there's an article "bicycle"
when the user navigates to the "bicycle" article's detail page
and clicks the "add to basket" button
then the article "bicycle" should be in their shopping basket

But how to make these human-readable scenarios into executable tests?

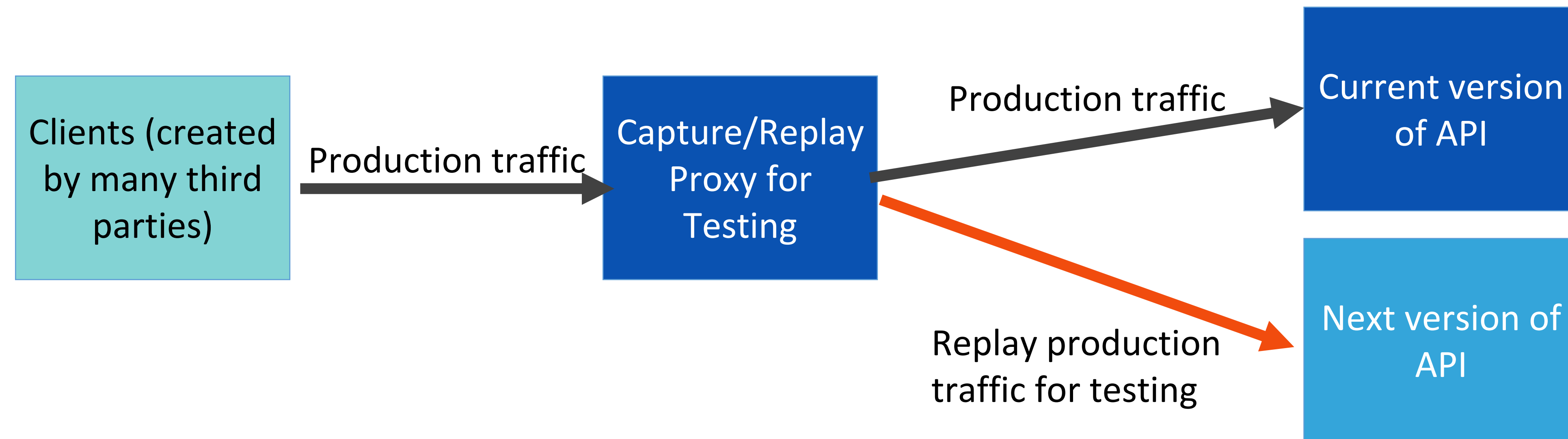
- Scenarios like the one above are readable by humans (e.g. customers)
- But they are not directly executable
- Many tools available to help fill this gap

Deployed systems create even more testing challenges

- Clients believe “how it is now is right”,
 - Not “how the API intended it to be is right”
 - Writing thorough test suite is even harder, less useful
 - What is a “breaking change”?
- Still: vital to detect breaking changes
- Examples:
 - Detailed layout of GUIs
 - Side-effects of APIs, particularly under corner-cases

Mock System-Level Components with Capture/Replay

- Record the API requests and responses that clients make
- Test new versions of the API by identifying requests that result in different responses ("breaking changes")

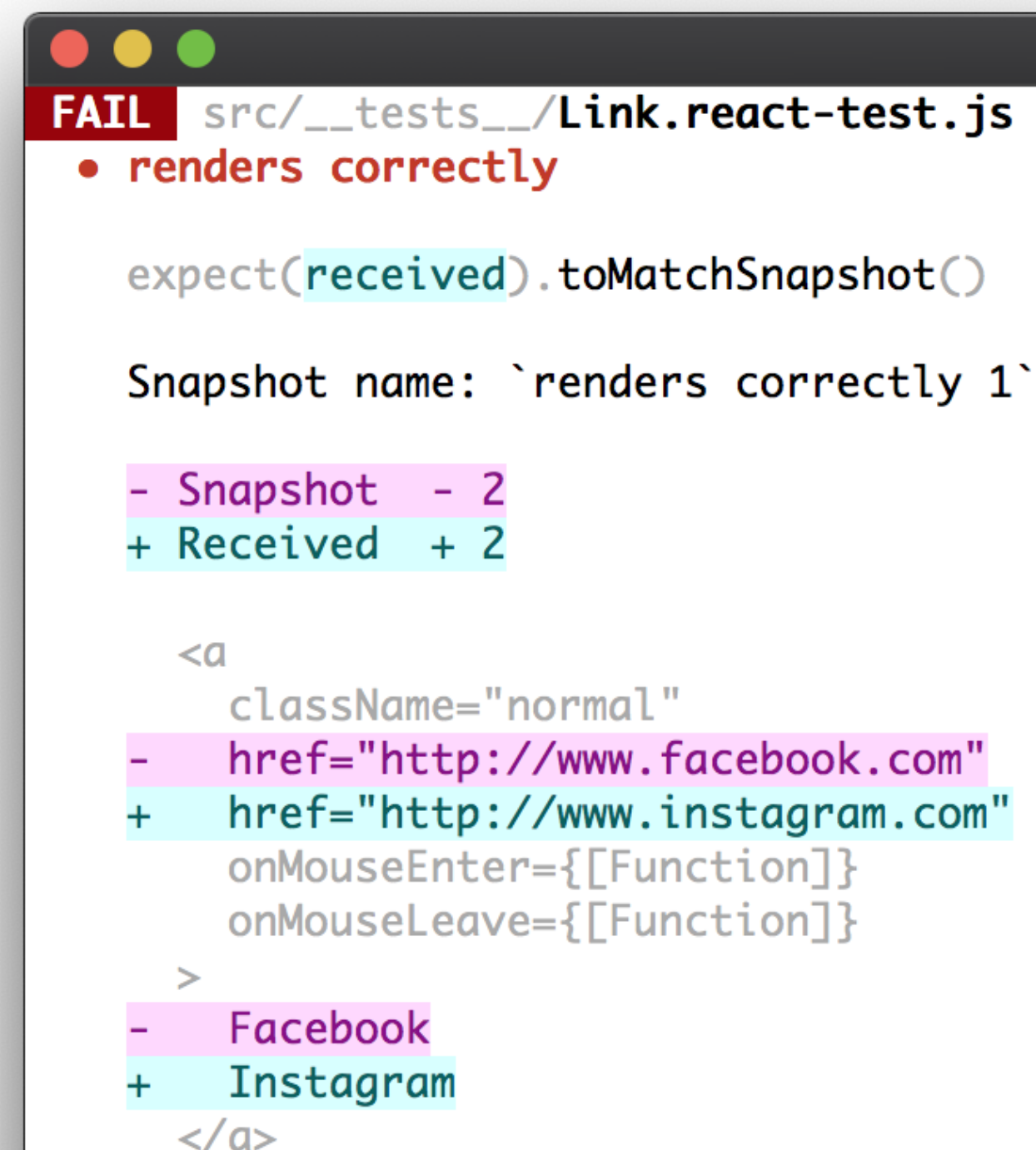


Snapshot Tests Can Detect GUI Changes

- The first time the test runs, it saves a "snapshot" of the rendered GUI
- Subsequent runs will fail if the snapshot changes

```
import renderer from 'react-test-renderer';
import Link from '../Link';

it('renders correctly', () => {
  const tree = renderer
    .create(<Link
page="http://www.facebook.com">Facebook</Link>)
    .toJSON();
  expect(tree).toMatchSnapshot();
});
```



```
FAIL src/__tests__/Link.react-test.js
  • renders correctly

expect(received).toMatchSnapshot()

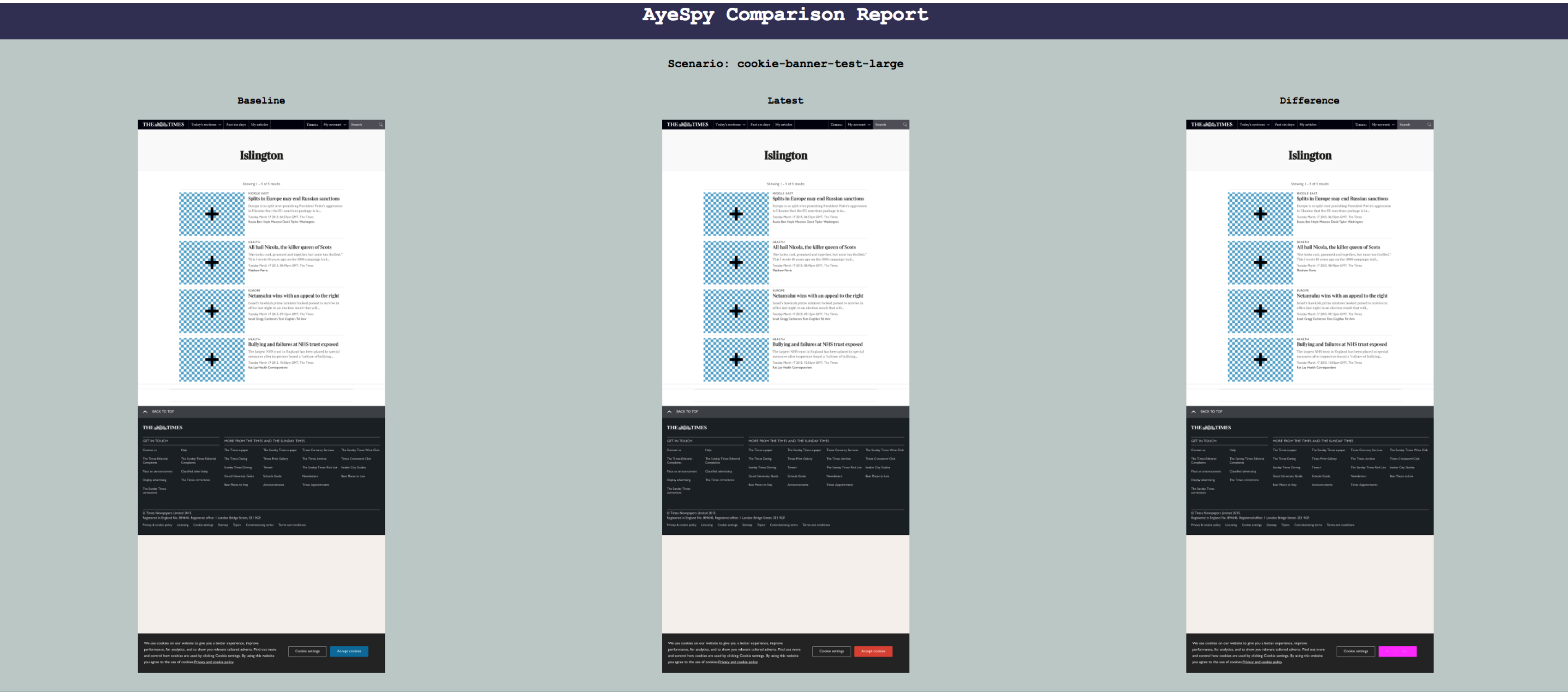
Snapshot name: `renders correctly 1`

- Snapshot - 2
+ Received + 2

<a
  className="normal"
- href="http://www.facebook.com"
+ href="http://www.instagram.com"
  onMouseEnter={[[Function]]}
  onMouseLeave={[[Function]]}
>
- Facebook
+ Instagram
</a>
```

Product Owners can Assess Visual Snapshot Tests

- Capture a visual snapshot of an application under a state
- If that snapshot changes, produce a visual report for manual sign-off



Learning Objectives for this Lesson

- You should now be prepared to:
 - Design test cases for code using fakes, mocks and spies
 - Explain why you might need a test double in your testing
 - Explain why you might need tests that are larger than unit tests
 - Explain how large, deployed systems lead to additional testing challenges